Rivertek RT-8700 Magnetic Flow Transmitter -- Technical Manual

PRODUCT OVERVIEW

The Rivertek RT-8700 is a state-of-the-art magnetic flow transmitter designed specifically for monitoring treated effluent flow in wastewater treatment plants. This device utilizes advanced electromagnetic technology to provide accurate and reliable flow measurements, ensuring that the treated effluent meets regulatory standards before being discharged into receiving water bodies. The RT-8700 is ideal for applications requiring continuous flow monitoring and can withstand harsh environmental conditions, making it a reliable choice for wastewater facilities.

TECHNICAL SPECIFICATIONS

- Flow Range: The Rivertek RT-8700 offers a flow measurement range from 0.5 to 10,000 gallons per minute (GPM), providing versatility for various applications within the wastewater treatment process.
- Accuracy: The device boasts an accuracy of ±0.5% of the measured value, ensuring precise flow measurement critical for compliance and reporting.
- Electrode Material: The electrodes are constructed of high-grade titanium, providing excellent corrosion resistance and longevity even in aggressive environments typical of wastewater treatment.
- Output Signal: The RT-8700 transmits data via a 4-20 mA output signal, compatible with most existing control systems. Additionally, it features a digital communication option using Modbus RTU for integration with modern SCADA systems.

CALIBRATION PROCEDURE

To ensure optimal performance and accuracy of the Rivertek RT-8700, periodic calibration is recommended. Follow these steps for proper calibration:

1. Preparation: Before starting the calibration process, ensure that the flow transmitter is installed correctly and that there are no obstructions in the flow path.

2. Equipment Required: You will need a calibrated flow meter for comparison, a multimeter for signal verification, and appropriate safety equipment.

3. Power On: Turn on the RT-8700 and allow it to warm up for at least 15 minutes for stable readings.

4. Flow Testing: Using the calibrated flow meter, establish a flow rate within the range of 10% to 100% of the RT-8700?s capacity. Record the flow rate indicated by both the RT-8700 and the calibrated flow meter.

5. Adjust Settings: Access the RT-8700's configuration menu through the user interface. Compare the readings from both devices. If any discrepancies exceed the acceptable limits (±0.5%), adjust the calibration settings accordingly.

6. Repeat: Perform this process for multiple flow rates (10%, 50%, and 100% of capacity) to ensure accuracy across the flow range.

7. Final Verification: After adjustments, verify the output signal using a multimeter to ensure it reflects the calibrated flow rates accurately.

8. Documentation: Record all calibration data, including flow rates, readings, and adjustments made, in the maintenance log for future reference.

ERROR CODES

The Rivertek RT-8700 features built-in diagnostic capabilities to assist in troubleshooting. Below are common error codes, their causes, and recommended resolutions:

- Error Code E01: "Signal Loss"
Cause: The flow signal is lost due to damaged wiring or connector issues.
Resolution: Check all electrical connections and wiring integrity. Replace damaged components if necessary.

- Error Code E02: "Electrode Fault"
Cause: A fault in the electrode circuitry or an issue with electrode integrity.
Resolution: Inspect electrodes for wear or corrosion. Clean or replace as needed.

- Error Code E03: "Calibration Error"
Cause: The device is out of calibration limits.
Resolution: Revisit the calibration procedure and adjust the settings as necessary.

- Error Code E04: "Flow Rate Exceeded"
Cause: The flow rate exceeds the maximum limit of the transmitter.
Resolution: Reduce the flow rate to remain within operational specifications. Investigate potential blockages upstream.

- Error Code E05: "Temperature Exceeded"
Cause: The ambient temperature exceeds the operational limits of the device.
Resolution: Check the installation environment. Ensure that the device is installed within the specified temperature range (-20°C to 60°C).

MAINTENANCE

Electrode Cleaning: Regular cleaning of the electrodes is crucial for maintaining the accuracy of the Rivertek RT-8700. Follow these guidelines for effective electrode maintenance:

1. Frequency: Inspect electrodes monthly and perform cleaning as needed, particularly in applications with high sediment or particulate content.

2. Cleaning Procedure:
- Power down the RT-8700 and ensure that the flow is stopped.
- Remove the electrode access cover using a suitable screwdriver.
- Carefully clean the electrodes with a soft cloth or brush to remove any buildup. For stubborn deposits, a mild detergent solution may be used.
- Rinse with clean water to remove any cleaning agents and ensure no residue remains.

3. Reassembly: Reattach the electrode access cover securely and power on the unit. Verify that there are no

error codes and that the device is functioning correctly.

4. Documentation: Log all maintenance activities in the device's maintenance record for tracking and future reference.

By following these guidelines, the Rivertek RT-8700 will provide reliable performance, ensuring compliance with wastewater discharge standards and effective monitoring of treated effluent flow.